Computer and Internet Use in the United States: 2021

American Community Survey Reports

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INTRODUCTION[1]

Access to the internet via computer or smartphone is an important part of many Americans' day-to-day lives. People use the internet for completing personal and professional tasks and for interacting with others across vast geographic distances. From browsing social media to paying bills online and countless other activities in between, access to the internet is nearly essential for many Americans. In recent years, the COVID-19 pandemic has further highlighted the importance of having reliable access to the internet, as businesses adopted remote telework policies, health care workers offered telehealth visits, and schools and universities switched to virtual instruction.

Given its importance, internet availability and utilization has been of high interest to academic researchers and policymakers alike. In the United States, household ownership of a computer and subscription to a broadband internet connection have increased over time. However, the magnitude of these changes has varied depending on important socioeconomic, racial and ethnic, and geographic characteristics. Data from the U.S. Census Bureau—which routinely surveys millions of households across the country on a variety of social and economic topics—provide a unique opportunity to study these differences as they develop and at multiple levels of geography.

[1] The U.S. Census Bureau reviewed this data product for unauthorized disclosure of confidential information and approved the disclosure avoidance practices applied to this release. Data Management System (DMS) number: P-7533841, Disclosure Review Board (DRB) approval number: CBDRB-FY24-SEHSD003-013.

What Is the American Community Survey?

The American Community Survey (ACS) is a nationwide survey designed to provide communities with reliable and timely demographic, social, economic, and housing data for the nation, states, congressional districts, counties, places, and other localities every year. It has an annual sample size of about 3.5 million addresses across the United States. The ACS releases 1-year estimates for geographies with populations of 65,000 people or more and 5-year estimates for all geographic areas down to the census tract and block group levels. For information on the ACS sample design and other topics, visit <www.census.gov/programs-surveys/acs/>.

This report explores data on computer and internet use in the United States from the 2021 American Community Survey (ACS), focusing on the demographic and geographic variations in these topics. This report focuses on the 2021 ACS, 1-year data, but also includes 2017–2021 ACS, 5-year data to provide estimates for smaller geographies and 1-year data from prior years to provide historical estimates related to computer and internet use. More information about the ACS can be found in the text box titled "What Is the American Community Survey?" The questions used in the 2021 American Community Survey are provided in Figure 1.

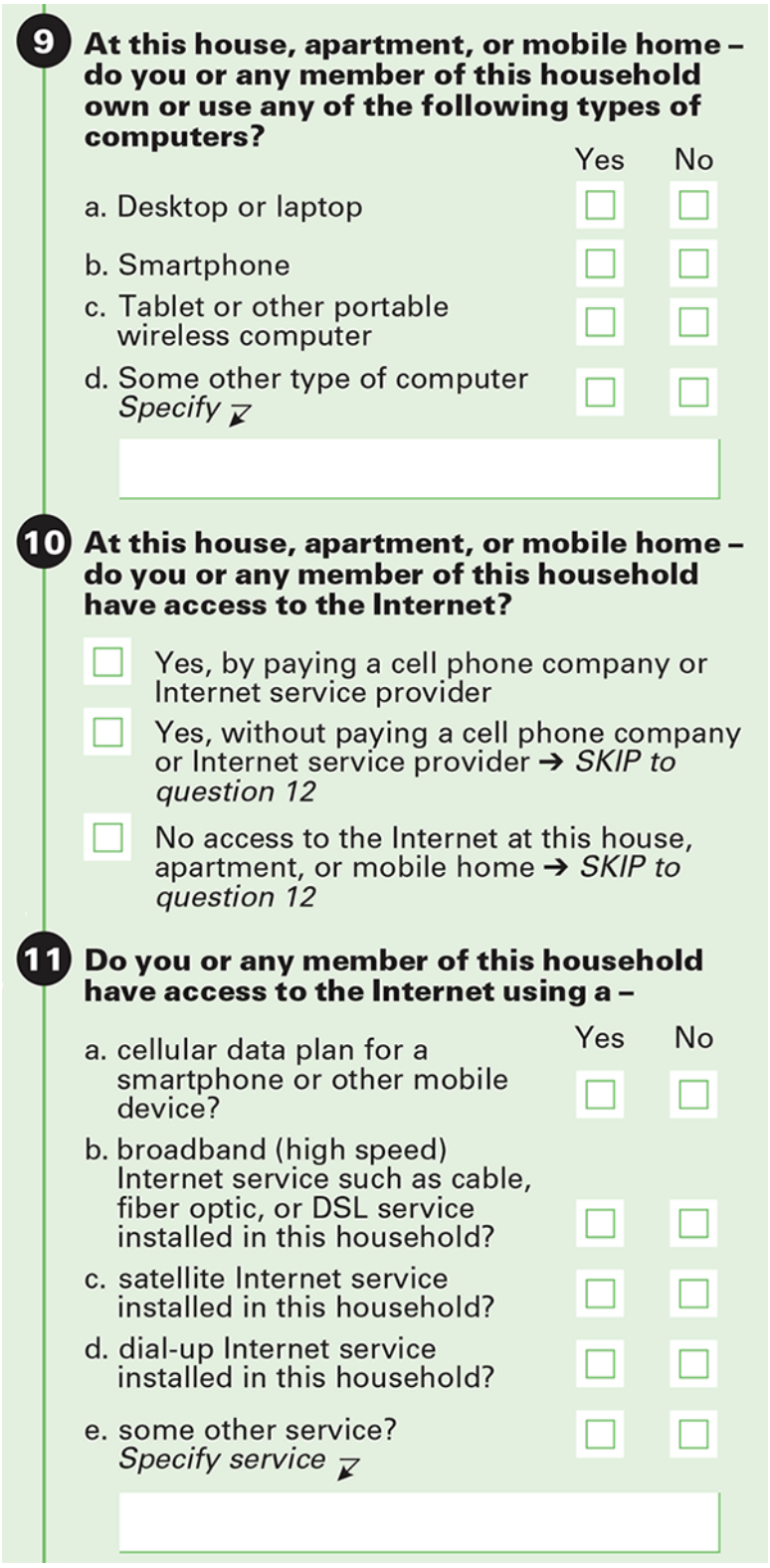
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HIGHLIGHTS FROM THE ACS

- Among all households in 2021, 95 percent had at least one type of computer, and 90 percent had a broadband internet subscription.[2, 3]
- Smartphones were the most common computing device, as 90 percent of households had at least one. Eighty-one percent of households had a desktop or laptop computer, followed by 64 percent of households that owned a tablet.
- Urban residents were more likely than rural residents to use computing devices (96 percent of urban households compared to 93 percent of rural households) and were more likely to have a broadband internet subscription (91 percent of urban households compared to 87 percent of rural households).
- Characteristics associated with lower internet subscription rates were a householder who rented rather than owned a home, householders with limited English proficiency, and households with at least one person with a disability.
- "Smartphone-only" households, which accessed the internet only through mobile broadband, were more likely to make $25,000 or less annually, be headed by someone 65 years and over, or have a Black or Hispanic householder.[4]

Figure 1.
ACS Questions on Computer and Internet Use: 2021



Source: U.S. Census Bureau, 2021 American Community Survey.

[2] The ACS considers all desktops, laptops, tablets, and smartphones as computers, along with selected computing technologies (such as smart home devices) and single board computers (such as Raspberry Pi and Arduino boards).

[3] Broadband refers to those who said they had at least one of the following types of internet subscriptions: cable, fiber optic, or DSL; a cellular data plan; satellite; a fixed wireless subscription; or other non-dial-up subscription types.

[4] In this report, non-Hispanic White alone, non-Hispanic Black alone, and non-Hispanic Asian alone groups are referred to as White, Black, and Asian, respectively.

Figure 2.
Percentage of Households With Computer and Internet Use: 2013 to 2021



Note: Dotted lines represent the ACS questionnaire change from 2015 to 2016 for data collection on computer and internet use. For more information, refer to <www.census.gov/programs-surveys/acs/technical-documentation/user-notes/2017-03.html>. No estimate is included for 2020 due to the impact of the COVID-19 pandemic on ACS estimates. For more information, refer to <https://www.census.gov/programs-surveys/acs/data/experimental-data.html>.

Source: U.S. Census Bureau, 2013–2021 American Community Survey, 1-year estimates.

TRENDS IN COMPUTER AND INTERNET USE OVER TIME

Figure 2 shows the percentage of households with a computer or internet subscription from 2013 to 2021. When the questions were first introduced to the survey in 2013, 84 percent of households had a computer of any type, and 74 percent subscribed to any type of internet. In 2016, the computer and internet questions were revised to improve responses to the questionnaire. One major change was the combination of DSL, cable, and fiber optic internet service into a single broadband category.[5] That year, household computer ownership was at 89 percent and subscription to the internet was at 82 percent. By 2021, 95 percent of households had a computer and 90 percent subscribed to the internet.

[5] For more information on questionnaire changes, visit <www.census.gov/programs-surveys/acs/technical-documentation/user-notes/2017-03.html>.

NATIONAL CHARACTERISTICS OF COMPUTER AND INTERNET USE: 2021

Table 1 shows computer and internet use by a variety of demographic, social, and geographic characteristics. Smartphones were the most common type of computing device, a trend that has been present since 2017. Nationally, 90 percent of households owned at least one smartphone, while 81 percent of households owned a desktop or laptop computer. Tablets trailed smartphones,

Table 1.

Computer and Internet Use for Households by Selected Characteristics: 2021

Household characteristics	Total households (in thousands)		Households with a computer								Households with an internet subscription			
			Any computer type		Desktop or laptop computer		Smartphone		Tablet		With any internet subscription		With a broadband subscription	
	Estimate	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]
Total households	**127,545**	**98**	**95.0**	**Z**	**80.5**	**0.1**	**90.0**	**Z**	**63.8**	**0.1**	**90.3**	**0.1**	**90.1**	**0.1**
Age of Householder														
15–34 years	24,258	66	98.9	Z	85.3	0.2	97.9	0.1	65.4	0.2	94.7	0.1	94.6	0.1
35–44 years	22,460	49	98.7	Z	87.3	0.2	97.3	0.1	75.3	0.2	95.4	0.1	95.4	0.1
45–64 years	46,540	55	96.8	0.1	82.6	0.1	93.1	0.1	66.9	0.1	92.3	0.1	92.2	0.1
65 years and older	34,280	56	87.5	0.1	69.8	0.2	75.4	0.1	50.9	0.2	81.2	0.1	80.7	0.1
Race and Hispanic Origin of Householder														
White alone, not Hispanic	82,172	65	95.0	Z	82.7	0.1	89.2	0.1	64.9	0.1	90.8	0.1	90.6	0.1
Black alone, not Hispanic	15,000	43	93.2	0.1	71.7	0.3	88.6	0.1	57.6	0.3	86.5	0.2	86.4	0.2
Asian alone, not Hispanic	6,386	31	98.0	0.1	89.7	0.2	95.5	0.1	74.1	0.3	95.0	0.2	94.9	0.2
Hispanic (of any race)	18,294	45	95.6	0.1	74.8	0.2	92.7	0.1	60.7	0.2	89.6	0.1	89.5	0.1
Age of Household Members														
Without members under 18	89,900	129	93.4	Z	77.2	0.1	87.0	0.1	57.2	0.1	87.9	0.1	87.7	0.1
With member(s) under 18	37,650	93	98.9	Z	88.3	0.1	97.3	Z	79.6	0.2	96.0	0.1	95.9	0.1
Limited English-Speaking Household[2]														
No	122,200	108	95.3	Z	81.6	0.1	90.2	Z	64.7	0.1	90.8	0.1	90.6	0.1
Yes	5,312	42	89.6	0.3	55.4	0.4	85.2	0.3	43.0	0.4	79.8	0.3	79.7	0.3
Households Where at Least One Member Has a Disability														
No	94,140	128	96.7	Z	83.6	0.1	92.7	Z	66.4	0.1	92.5	0.1	92.4	0.1
Yes	33,410	94	90.5	0.1	71.7	0.1	82.5	0.1	56.6	0.2	84.1	0.1	83.8	0.1
Housing Tenure														
Owner occupied	83,397	188	95.8	Z	84.7	0.1	90.3	Z	68.6	0.1	92.2	0.1	92.0	0.1
Renter occupied	44,148	126	93.6	0.1	72.6	0.1	89.5	0.1	54.6	0.1	86.7	0.1	86.6	0.1

Footnotes provided at end of table.

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	Estimate	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]	Percent	Margin of error (±)[1]
Urban and Rural by Region														
Urban area	102,611	78	95.5	Z	81.5	0.1	90.7	0.1	64.6	0.1	91.1	0.1	91.0	0.1
Rural area	24,933	122	93.1	0.1	76.4	0.1	87.1	0.1	60.4	0.2	86.9	0.1	86.6	0.1
Northeast	22,419	25	94.5	0.1	81.4	0.1	88.2	0.1	64.8	0.2	90.8	0.1	90.6	0.1
Urban area	19,094	29	94.6	0.1	81.5	0.1	88.7	0.1	65.0	0.2	91.0	0.1	90.9	0.1
Rural area	3,325	21	93.4	0.2	81.2	0.3	85.1	0.2	63.7	0.4	89.6	0.2	89.2	0.2
Midwest	27,682	36	94.4	0.1	79.0	0.2	88.8	0.1	62.4	0.2	89.7	0.1	89.5	0.1
Urban area	21,132	44	94.9	0.1	79.8	0.2	89.6	0.1	63.2	0.2	90.5	0.1	90.3	0.1
Rural area	6,550	60	92.9	0.1	76.5	0.2	86.4	0.2	60.0	0.2	87.2	0.2	86.8	0.2
South	48,886	57	94.9	0.1	78.5	0.1	90.3	0.1	62.3	0.2	89.2	0.1	89.0	0.1
Urban area	37,051	47	95.6	0.1	80.1	0.1	91.3	0.1	63.5	0.2	90.4	0.1	90.3	0.1
Rural area	11,834	47	92.6	0.1	73.3	0.2	87.5	0.2	58.5	0.3	85.3	0.2	85.1	0.2
West	28,559	26	96.4	0.1	84.7	0.1	92.0	0.1	66.9	0.1	92.4	0.1	92.2	0.1
Urban area	25,335	35	96.6	0.1	85.0	0.1	92.4	0.1	67.2	0.2	92.8	0.1	92.6	0.1
Rural area	3,224	27	95.1	0.2	82.9	0.4	89.3	0.3	64.7	0.5	89.5	0.3	89.2	0.3
Household Income														
Less than $25,000	22,810	89	85.1	0.1	56.3	0.2	76.7	0.2	40.6	0.2	75.2	0.2	74.9	0.2
$25,000–$49,999	24,800	75	93.4	0.1	71.9	0.2	85.9	0.1	52.0	0.2	87.0	0.1	86.8	0.1
$50,000–$99,999	37,970	114	97.6	Z	85.1	0.1	93.0	0.1	66.0	0.1	93.8	0.1	93.6	0.1
$100,000–$149,999	20,420	80	98.9	Z	92.7	0.1	96.3	0.1	77.4	0.2	96.7	0.1	96.6	0.1
$150,000 and more	21,550	72	99.3	Z	96.3	0.1	97.6	0.1	85.3	0.1	97.8	0.1	97.8	0.1
Total households with householders 25 years and over	**122,800**	**92**	**94.9**	**Z**	**80.4**	**0.1**	**89.7**	**Z**	**64.2**	**0.1**	**90.2**	**0.1**	**90.1**	**0.1**
Educational Attainment of Householder														
Less than high school graduate	10,890	70	83.8	0.2	51.3	0.4	77.7	0.2	41.4	0.3	74.3	0.3	74.2	0.3
High school graduate (includes equivalency)	28,570	85	90.7	0.1	66.3	0.2	82.9	0.1	51.2	0.2	83.8	0.1	83.5	0.1
Some college or associate degree	36,210	102	96.8	0.1	82.9	0.1	91.4	0.1	65.5	0.2	92.3	0.1	92.1	0.1
Bachelor's degree or higher	47,090	153	98.6	Z	93.9	0.1	95.3	Z	76.2	0.2	96.3	0.1	96.2	0.1

Z Margin of error is less than 0.1.

[1] Data are based on a sample and are subject to sampling variability. A margin of error is a measure of an estimate's variability. The larger the margin of error in relation to the size of the estimate, the less reliable the estimate. This number when added to or subtracted from the estimate forms the 90 percent confidence interval.

[2] A "limited English-speaking household" is one in which there is no one aged 14 and over who speaks English at home or who speaks English "very well."

Source: U.S. Census Bureau, 2021 American Community Survey, 1-year estimates.

desktops, and laptops, as 64 percent of households owned at least one—an increase of only 1 percent since 2018. Most households had at least some sort of internet subscription (90 percent), and in most cases this subscription was to a broadband service. Over one-half of all households in the United States (56 percent) owned all three device types (laptop or desktop, smartphone, and tablet) as well as had a broadband internet connection.

HOUSEHOLDER DEMOGRAPHICS

Households with older householders tended to have lower levels of both computer ownership and broadband internet subscription than those in younger age groups.[6] Eighty-eight percent of households with a householder aged 65 and over owned some type of computer, while all other age groups had ownership rates above 96 percent. Smartphones were owned in roughly 75 percent of households with a householder aged 65 and over, and in roughly 98 percent of those with householders under 35 years. Similarly, only 81 percent of households with a householder aged 65 and over had a broadband subscription, compared to 92 percent for those between the ages of 45 and 64, the group with the next lowest subscription rate.

Computer and internet use also varied by race and Hispanic origin of the householder. Households with an Asian householder had the highest computer ownership rates across all device types and were the racial group most likely to have a broadband internet subscription. In contrast, households with a Black householder were the least likely to own a computer of any type or to have a broadband subscription. Hispanic householders had device ownership levels below Asian and White householders and above Black householders for all computer types, except for smartphones, where they had the second-highest ownership rate (93 percent for Hispanic householders compared to 96 percent for Asian householders).

Differences in smartphone ownership across households headed by each race and Hispanic origin group were also smaller than differences in desktop or laptop ownership or broadband subscription. For example, the difference between Asian and Black householders in smartphone ownership was 7 percentage points, while the gap for laptop or desktop ownership was 18 points, and 17 points for tablet ownership.

Household Structure

Several factors involving household structure were associated with computer ownership and internet subscription. Households with children under 18 years were more likely to have any type of computer (99 percent) and a broadband internet subscription (96 percent) than households without children. Limited English-speaking households trailed other households for both computer ownership and broadband subscription rates.[7] Additionally, householders who owned their home were more likely to own a computer (96 percent) and subscribe to the internet (92 percent) compared to householders who rented (94 percent and 87 percent, respectively).

Ninety-one percent of households in which at least one member had a physical or mental disability owned any type of computer, while 97 percent of households with no members with a disability owned a computer. These households also had a broadband subscription 84 percent of the time, compared to 92 percent of households without members with a disability.

Socioeconomic Context

Household income and educational attainment of the householder, key indicators of socioeconomic status, were closely linked to computer ownership and internet subscription. Over 99 percent of households earning $150,000 or more annually had some sort of computing device, compared to 85 percent of households making less than $25,000. In addition, 99 percent of households in which the householder had a bachelor's degree or higher had some sort of computer, compared to 84 percent of households in which the householder was less than a high school graduate. Broadband subscription rates also differed by at least 22 percentage points across both income and education at the highest and lowest levels.

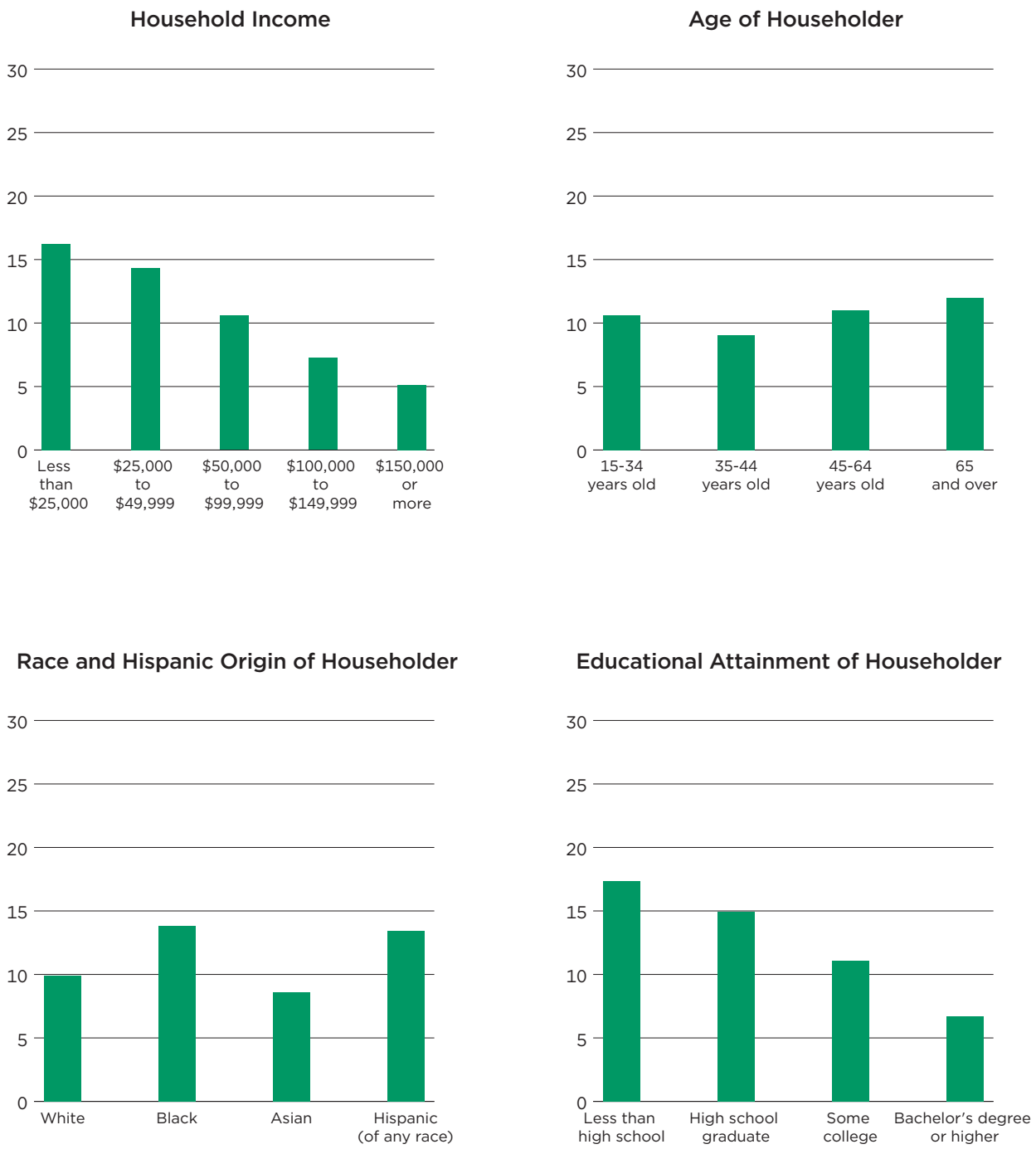
Households with high income earners and high educational attainment were also more likely to own multiple computing devices (not shown in Table 1). Eighty-two percent of householders earning $150,000 or more annually had a desktop or laptop, smartphone, tablet, and a broadband internet connection at home, compared to 29 percent of householders making less than $25,000. Similarly, 72 percent of households in which the householder had a bachelor's degree or higher had all computing device types and a broadband subscription, compared to 30 percent of households in which the householder did not have a high school diploma.

[6] The Census Bureau defines a householder as the person (or one of the people) in whose name the housing unit is owned or rented (maintained), or if there is no such person, any adult member, excluding roomers, boarders, or paid employees. If the house is owned or rented jointly by a married couple, the householder may be either spouse.

[7] A "limited English-speaking household" is one in which there is no one aged 14 and over who speaks English at home or who speaks English "very well."

Figure 3.

Percentage of "Smartphone-Only" Households by Selected Characteristics: 2021



Source: U.S. Census Bureau, 2021 American Community Survey, 1-year estimates.

Smartphone-Only Households

Eleven percent of households accessed the internet only through a cellular data plan for a smartphone or other mobile device instead of a fixed broadband connection. These "smartphone-only" households were more likely to make $25,000 or less annually, be headed by someone 65 years and over, or have a Black or Hispanic householder (Figure 3). Specifically, 16 percent of households earning $25,000 or less were smartphone-only, compared to 5 percent of households earning $150,000 or more; 12 percent of households with a householder aged 65 and over were smartphone-only, compared to 9 percent of 35- to 44-year-old householders; 14 percent of Black households and 13 percent of Hispanic households were smartphone-only, compared to 9 percent of Asian households; and 17 percent of households headed by a householder with less than a high school diploma were smartphone-only, compared to 7 percent of households headed by a householder with a bachelor's degree or higher educational attainment.

Geographic Context

Rates of computer ownership and internet use varied across regions and geographic areas. Urban households owned all types of computers and reported broadband subscriptions more frequently than those in rural areas.[8] This was true for all device types in almost every geographic region of the United States except in the Northeast, where there was no significant difference between rural and urban households in desktop or laptop ownership. Overall, households in the West had higher rates of broadband subscription and computer ownership for all device types than households in all other geographic regions.

In most states, urban households owned computers more often than rural households. However, there were 12 states with no significant difference between rural and urban areas, and two states in the Northeast (Connecticut and Massachusetts) where rural households had higher rates of computer ownership compared to urban households.[9]

Broadband Internet Access Across States, Counties, and Tribal Areas

Broadband subscription rates also varied across states and the urban and rural areas within each state. Colorado, Utah, and Washington were the states with the highest broadband subscription rates, while Mississippi, New Mexico, and West Virginia ranked lowest. In most states, urban areas had higher broadband subscription rates than rural areas. Figure 4 illustrates the urban–rural broadband disparity by state. Appendix Table A-1 provides each state's rural, urban, and overall broadband rates. The urban–rural gap in connectivity has also narrowed or closed in several places. At the national level, broadband subscription in rural households trailed urban households by 4 percentage points in 2021. This gap is 2 percentage points smaller than in 2016. Additionally, ten states showed no significant difference in broadband subscription rates between rural and urban areas.

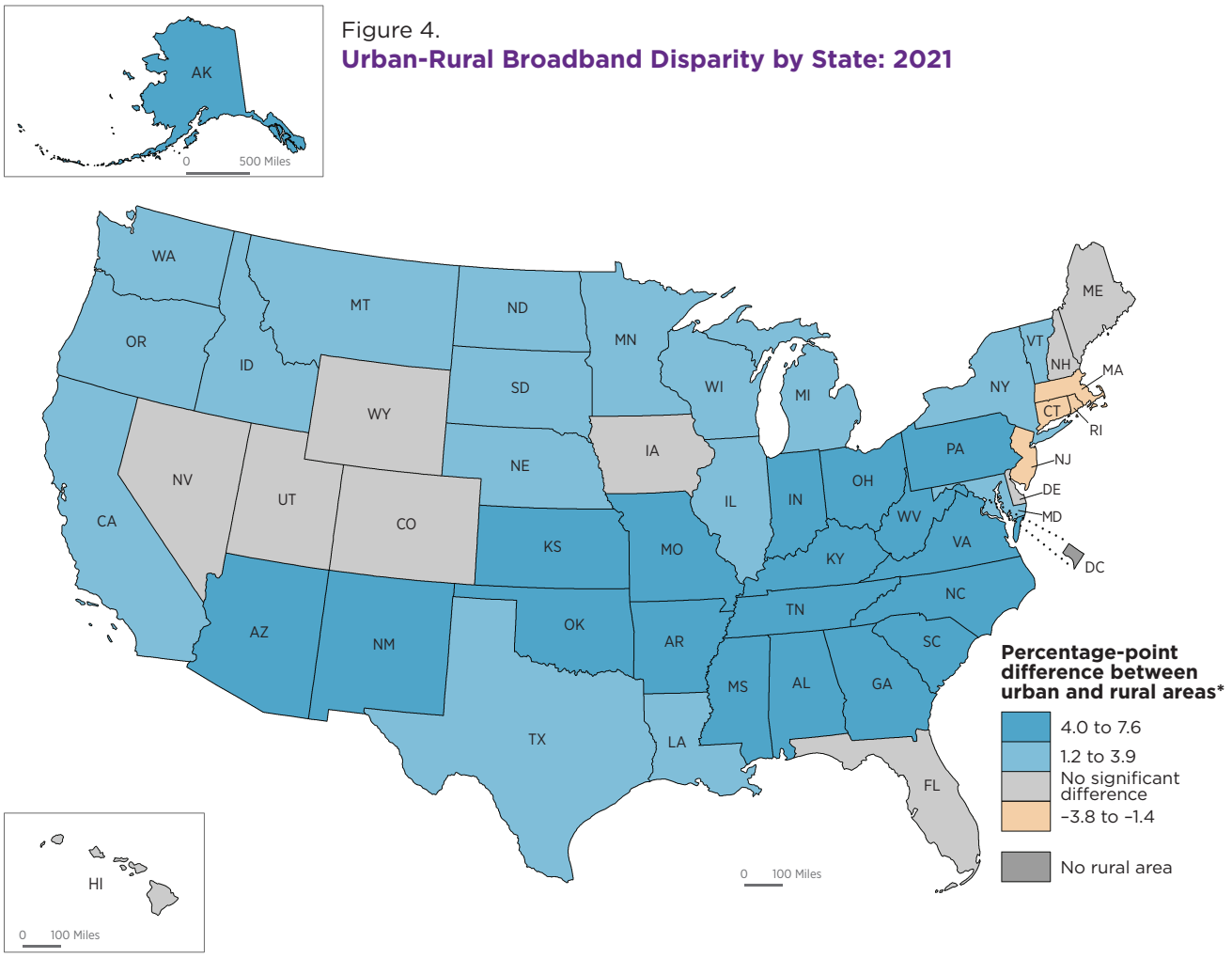
In a few states in the Northeast, rural broadband subscription even surpassed urban broadband subscription. Rural households' subscription rates were higher by 4 percentage points in Rhode Island, 3 points in Connecticut, 2 points in New Jersey, and 1 point in Massachusetts. In contrast, Virginia, New Mexico, and Alabama showed the largest urban–rural disparity, with all three states showing higher subscription rates in urban areas. Urban broadband rates exceeded rural broadband rates by 8 percentage points in Virginia and 7 percentage points in both New Mexico and Alabama.

Figure 5 presents broadband subscription rates by county using the 5-year, 2017–2021 ACS data. County broadband rates are grouped by the four highest 10-percent intervals (marked at 90 percent, 80 percent, 70 percent, and 60 percent) as well as households with less than 60 percent broadband. This map helps illustrate some origins of the urban–rural divisions presented at the state level in Figure 4. For example, counties in Virginia adjacent to Washington, DC, and Richmond had 90 percent or more household broadband subscription rates, while several counties in the southern and western parts of the state had less than 70 percent household broadband. Counties with the lowest household broadband rates (below 60 percent) were found in 17 states, with ten of these states located in the South, five in the West, and two in the Midwest. The number of counties with broadband subscription rates below 60 percent decreased from 291 counties in 2014–2018 to 66 counties in 2017–2021. Of the 66

[8] The Census Bureau defines urban areas as territories that encompass at least 2,500 people, at least 1,500 of whom must reside outside institutional group quarters. All other areas are defined as rural. Rural and urban classifications are updated after each decennial census. This report uses the urban-rural delineations defined by the 2010 Census. As a result, data for urban and rural areas from the ACS do not necessarily reflect the results of ongoing urbanization.

[9] The 12 states with no significant difference between urban and rural areas for computer ownership were Delaware, Hawaii, Idaho, Maine, Montana, Nevada, New Hampshire, New Jersey, North Dakota, Rhode Island, Utah, and Wyoming.

Figure 4.
Urban-Rural Broadband Disparity by State: 2021



* Percentage-point difference between urban and rural areas is significant at the 90 percent confidence level.

Source: U.S. Census Bureau, 2021 American Community Survey, 1-year estimates.

counties that fall into this category, 49 were in the South, 15 of which were in Georgia.

Compared to the 90 percent national average, 84 percent of households in tribal areas had broadband internet.[10] American Indian and Alaska Native households living on tribal lands had one of the lowest broadband subscription rates in the United States, at 71 percent. Since 2016, the divide between tribal and non-tribal broadband has decreased from a gap of 10 percentage points in 2016 to 6 points in 2021. Tribal areas also had an urban–rural gap in connectivity, with 88 percent of tribal households in urban areas subscribing to broadband internet, compared to 80 percent of tribal households in rural areas. For reference, 45 percent of tribal areas are rural, compared to 19 percent of non-tribal areas.

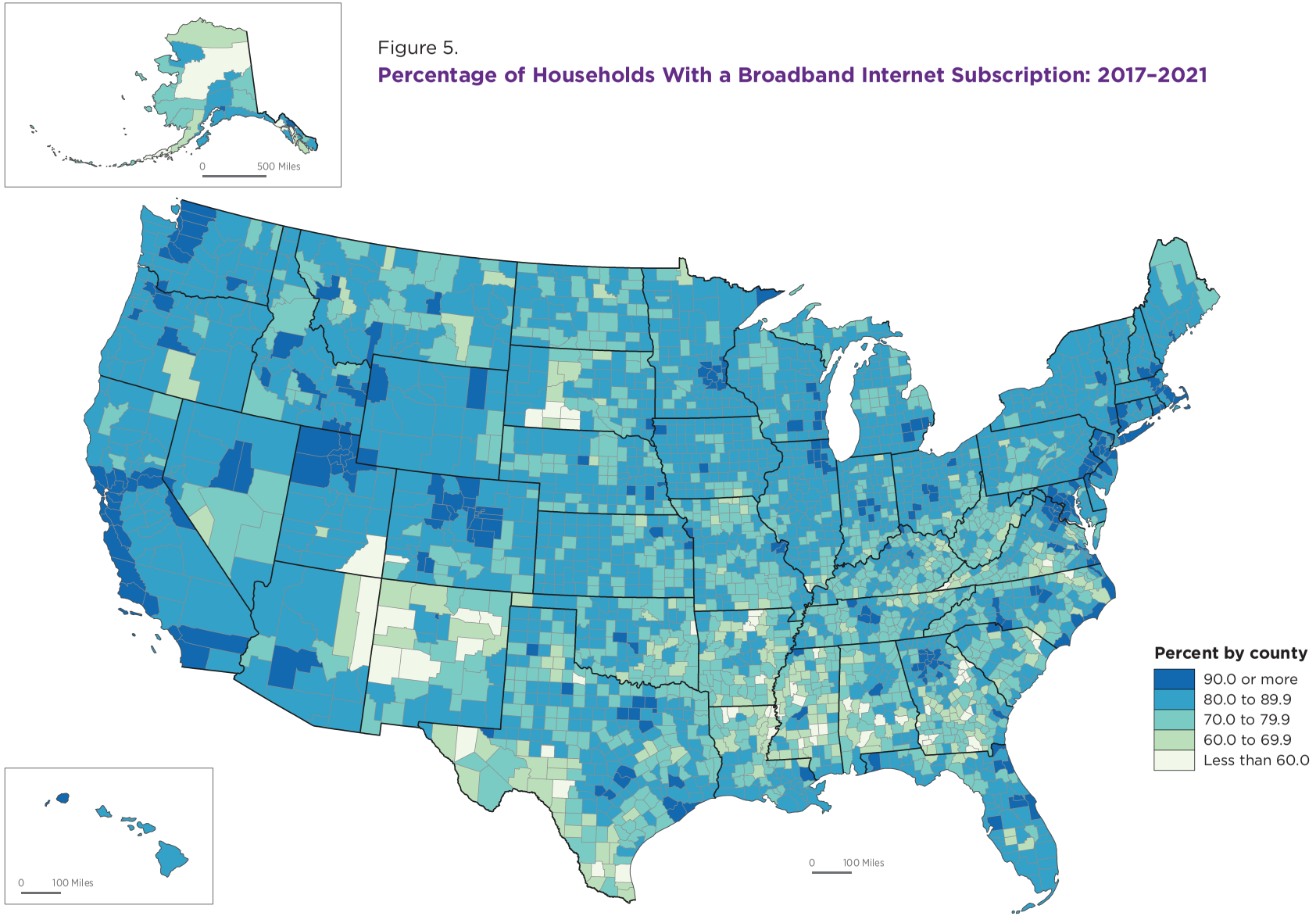
Not all tribal areas experience unequal access to broadband internet. Tribal residents in six states (Delaware, Florida, Georgia, Hawaii, Washington, and Wisconsin) had broadband rates at or above the national average of 90 percent.[11] Tribal residents in three states (Arizona, Connecticut, and New Mexico) had among the lowest rates, with 70 percent or less using broadband internet. However, over one-quarter (27 percent) of American Indian and Alaska Native households on tribal lands lived in states with the lowest tribal broadband rates, while only 4 percent lived in states with the highest tribal broadband rates.

[10] Tribal areas include all American Indian and Alaska Native Areas as defined by the ACS. They include Hawaiian homelands, Alaska Native village statistical areas, American Indian reservations and rancherias, and others. For more information, visit <https://www2.census.gov/geo/pdfs/reference/GARM/Ch5GARM.pdf>.

[11] Fifteen states were removed from consideration due to not having any federally recognized tribal areas or having too few sampled cases.

Figure 5.

Percentage of Households With a Broadband Internet Subscription: 2017–2021



Source: U.S. Census Bureau, 2017–2021 American Community Survey, 5-year estimates.

SUMMARY

This report highlights important computer and internet use findings for the United States from the 2021 American Community Survey. The data show continued positive trends for both computer ownership and internet subscription over time. However, not all households were equal in their rates of access, as substantial variation existed along socioeconomic, racial and ethnic, and geographic characteristics. Households with older householders, those with lower levels of income or education, and those without children tended to have lower levels of computer ownership and broadband internet access. Rural households were also less likely to own computers or access the internet except in a few states.

This report also highlights the importance of tracking the types of computers and internet subscriptions used in households. Smartphone ownership has exceeded desktop computer and laptop ownership since 2017, reflecting the growing importance of mobile access to the internet. Furthermore, certain households are "smartphone-only" and rely on mobile broadband to access the internet, which may affect the types of tasks these households can accomplish online. As technology continues to change, the Census Bureau will continue to measure computer and internet use throughout the country and its many communities.

SOURCE AND ACCURACY

The data presented in this report are from the American Community Survey (ACS). The analyses use data from 2013 to 2021 and are based on samples interviewed from January 1 through December 31 of each year. For example, the 2021 data are based on a sample interviewed from January 1, 2021, through December 31, 2021. The estimates based on the sample from each year describe the average values of person, household, and housing unit characteristics over the year of collection. Sampling error is defined as the uncertainty between an estimate based on a sample and the corresponding value that would be obtained if the estimate were based on the entire population. Measures of sampling error are provided in the form of margins of error for key estimates included in this report. All comparative statements for the ACS in this report have undergone statistical testing, and comparisons are significant at the 90 percent level, unless otherwise noted.

In addition to sampling error, nonsampling error may be introduced during any of the operations used to collect and process survey data such as editing, reviewing, or keying data from questionnaires. More information on sampling and estimation methods, confidentiality protection, and sampling and nonsampling errors can be found from the ACS, 1-year Accuracy of the Data documents for 2013 through 2021, located at <www.census.gov/programs-surveys/acs/technical-documentation/code-lists.html>.

This report also makes use of 2017–2021 ACS, 5-year data. These data are obtained by pooling data from each of the 5 years and then reweighting these data using similar processes to the approach with single-year data. These data reflect the average value over the entire period and should not be equated with a single-period estimate for a given geography. More information can be found at <https://www2.census.gov/programs-surveys/acs/tech_docs/accuracy/MultiyearACSAccuracyofData2021.pdf>.

Multiple changes were made in 2016 to ACS questions on computer use, internet access, and type of internet subscription. Changes include improving the measurement of internet subscriptions and cellular data plans among households with smartphones, as well as keeping up with rapid changes in the types of computing devices available and the terminology used to describe them. Because of these changes, caution should be used when comparing the estimates for 2016 and later to those from previous years, since changes may be due to the revised wording and improved measurement of the survey rather than a change in computer and internet use. More information can be found at <www.census.gov/programs-surveys/acs/technical-documentation/user-notes/2017-03.html>.

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Appendix Table 1.

Percentage of Households With a Broadband Internet Subscription by Rural and Urban Residence: 2021

Geographic area	All areas		Urban areas		Rural areas	
	Percent	Margin of error (±)	Percent	Margin of error (±)	Percent	Margin of error (±)
Utah	93.5	0.4	93.5	0.5	93.1	0.9
Washington	93.5	0.2	93.9	0.3	90.9	0.6
Colorado	93.0	0.3	93.1	0.3	92.5	0.7
California	92.9	0.1	93.1	0.1	89.8	0.5
New Hampshire	92.3	0.5	92.5	0.7	92.1	0.8
Massachusetts	92.2	0.3	92.1	0.3	93.5	0.9
Connecticut	92.2	0.4	91.8	0.4	94.8	0.9
Delaware	91.9	0.8	92.0	0.9	91.1	1.9
New Jersey	91.9	0.3	91.7	0.3	94.2	0.9
Maryland	91.8	0.3	92.2	0.3	89.1	0.9
Oregon	91.6	0.4	92.1	0.4	89.6	1.0
Arizona	91.4	0.3	92.0	0.3	86.6	1.1
Hawaii	91.3	0.7	91.5	0.7	89.7	2.6
Minnesota	91.3	0.3	92.0	0.3	89.2	0.4
Rhode Island	90.8	0.8	90.4	0.9	94.3	2.2
Alaska	90.7	0.9	92.6	1.0	86.8	1.5
Virginia	90.6	0.3	92.5	0.3	85.0	0.7
Nevada	90.5	0.5	90.6	0.5	89.5	2.1
Florida	90.5	0.2	90.5	0.2	90.0	0.6
Idaho	90.5	0.6	90.9	0.7	89.4	0.9
New York	90.4	0.2	90.6	0.2	89.4	0.4
Wyoming	90.3	1.1	89.8	1.5	91.2	1.6
Texas	90.1	0.2	90.5	0.2	88.3	0.4
District of Columbia	90.1	0.9	90.1	0.9	X	X
United States	**90.1**	**0.1**	**91.0**	**0.1**	**86.6**	**0.1**
Michigan	90.0	0.2	90.8	0.2	87.5	0.4
Georgia	90.0	0.3	91.5	0.3	85.3	0.7
Kansas	89.8	0.4	90.9	0.5	86.7	0.8
Illinois	89.8	0.2	90.2	0.3	86.3	0.6
Nebraska	89.8	0.5	90.7	0.6	87.2	0.8
Maine	89.8	0.7	89.7	1.1	89.8	0.8
Wisconsin	89.6	0.3	90.5	0.4	87.6	0.4
Indiana	89.3	0.3	90.4	0.4	86.3	0.7
Vermont	89.2	0.9	91.1	1.5	88.0	1.1
North Carolina	88.9	0.3	90.7	0.3	85.2	0.5
Ohio	88.9	0.3	89.7	0.3	85.7	0.5
Pennsylvania	88.8	0.2	89.6	0.3	85.6	0.5
Montana	88.8	0.7	90.0	1.1	87.2	1.0
Missouri	88.5	0.3	90.1	0.4	84.2	0.7
Iowa	88.3	0.4	88.4	0.6	88.3	0.6
North Dakota	88.1	0.8	88.8	1.2	86.9	1.2
Tennessee	88.0	0.4	89.8	0.4	84.3	0.6
South Dakota	88.0	0.7	89.4	1.0	85.9	1.0
Oklahoma	87.8	0.3	89.3	0.4	84.6	0.5
South Carolina	87.8	0.5	89.8	0.5	83.5	0.9
Kentucky	87.1	0.4	89.1	0.5	83.9	0.7
Arkansas	85.5	0.6	87.6	0.8	82.6	0.8
Louisiana	85.4	0.5	86.3	0.5	82.8	1.0
Alabama	85.0	0.5	87.6	0.5	81.1	0.8
West Virginia	85.0	0.7	87.2	0.9	82.5	1.2
New Mexico	84.3	0.7	85.9	0.9	78.6	1.4
Mississippi	81.8	0.7	84.3	1.0	79.4	0.9

X Not applicable. The District of Columbia has no area that is classified as rural.
Source: U.S. Census Bureau, 2021 American Community Survey, 1-year estimates.